

Familiar Faces

Industry-Integrated AI Systems Synthesis

A Human-Supervised Care-Transition Outreach Research Prototype

Prepared by: Grant Collings

Udacity AI for Industry Applications Capstone

July 28, 2026

EDUCATIONAL DEIDENTIFIED-DATA RESEARCH PROTOTYPE

Not approved for clinical care, operational deployment, eligibility decisions, or autonomous patient contact.

Executive Summary

Familiar Faces explores how a community paramedicine or care-transition team might use limited AI support to organize human review after hospitalization. The revised prototype implements three connected domains: descriptive analysis of real, deidentified hospital encounters; supervised machine learning for an advisory 30-day-readmission signal; and genuine large-language-model inference for structured, nonclinical review drafts. A validated fairness layer, deterministic output checks, and mandatory human review govern those components. The source is the public UCI Diabetes 130-US Hospitals dataset, which contains 101,766 encounters from 1999 through 2008 (Clare et al., 2014). After excluding hospice or expired dispositions and retaining one eligible encounter per patient, the analytic cohort contains 69,990 encounters.

Four-fold cross-validation within the training data selected histogram gradient boosting before the final test data were examined. On the untouched test partition, it achieved ROC AUC 0.641 (95% bootstrap CI [0.626, 0.656]) and average precision 0.171 (95% bootstrap CI [0.156, 0.186]). Those results are moderate and support only a review-queue experiment. The fairness rule produced 0 review triggers. In 200 independent-null simulations its familywise false-positive rate was 0.0%; in 200 simulations with an injected 0.15 flag-rate disparity, detection was 100.0%. Live structured inference generated 5 case drafts, and 5 passed the separate safety validator. Passing does not authorize action; the prototype preserves an explicit human terminal boundary.

1. Industry Problem and Scope

Emergency response resolves immediate threats, but repeated emergency use and hospital readmission may also reflect needs that a single encounter cannot address. Community paramedicine and care-transition teams can coordinate follow-up, verify discharge understanding, and connect people with authorized resources. The responsible AI opportunity is therefore narrow: help staff sort a review queue and prepare consistent questions. It is not to diagnose, predict who deserves care, or replace professional judgment. Readmission is used because it is an observed outcome in the source data, not because it fully represents outreach need or likely benefit.

The system estimates whether an encounter was followed by readmission within 30 days. Staff would have to interpret that signal alongside consent, current circumstances, program criteria, and information unavailable to the model. No component may determine eligibility, recommend treatment or medication changes, or contact anyone. This boundary recognizes that prioritization can itself affect access and attention. The appropriate capstone claim is technical and governance feasibility—not clinical effectiveness.

The intended reviewer is an authorized professional, not the patient and not an automated outreach service. The queue would provide a starting point for review, while staff remain responsible for deciding whether a record is current, whether contact is permitted, and whether the program has an appropriate resource. A low score must never block review, and a high score must never compel contact. This asymmetry is deliberate because the cost of mistaken exclusion differs from the cost of asking a reviewer to inspect another record.

2. Cross-Domain Integration and Rationale

Data Workflow. Earlier data work established the importance of explicit cohort definitions, traceable transformations, schema validation, and reproducible outputs. Those practices now operate on a real outcome rather than a formula-generated synthetic label. The pipeline documents exclusions, retains the first eligible encounter per patient, and writes a processed cohort plus data dictionary. This directly removes circular evaluation: neither predictors nor labels are generated from the model being evaluated.

Machine Learning Workflow. The NEMSIS modeling work informed fixed seeds, patient-independent partitioning and comparison of interpretable linear and nonlinear models. Four transparent utilization features were derived from

the source counts. Logistic regression, random forest, and histogram gradient boosting were tuned using four-fold cross-validation inside the training data. Average precision was the primary selection measure because only about nine percent of held-out encounters have the positive outcome; ROC AUC was secondary. Only the selected configuration was evaluated on the untouched test set, and bootstrap intervals quantify uncertainty. Threshold metrics remain visible so that no single score hides the operational tradeoff. Feature importance explains model behavior, not causality.

Generative AI. The earlier narrative-generation work showed that fluent text can invent unsupported detail and sound more authoritative than its evidence. This synthesis therefore implements a genuine but constrained LLM component. Five records are converted to demographics-free, deidentified inputs containing only utilization counts, encounter counts, and the model probability. The OpenAI Responses API returns a defined schema containing a summary, three verification questions, conditional coordination options, limitations, and a human-review notice. A deterministic validator then checks every output for protected-demographic language, clinical or prescribing language, dose changes, autonomous eligibility or contact, and required warning phrases. Structured output improves schema reliability but does not make generated content true or safe by itself (OpenAI, 2026).

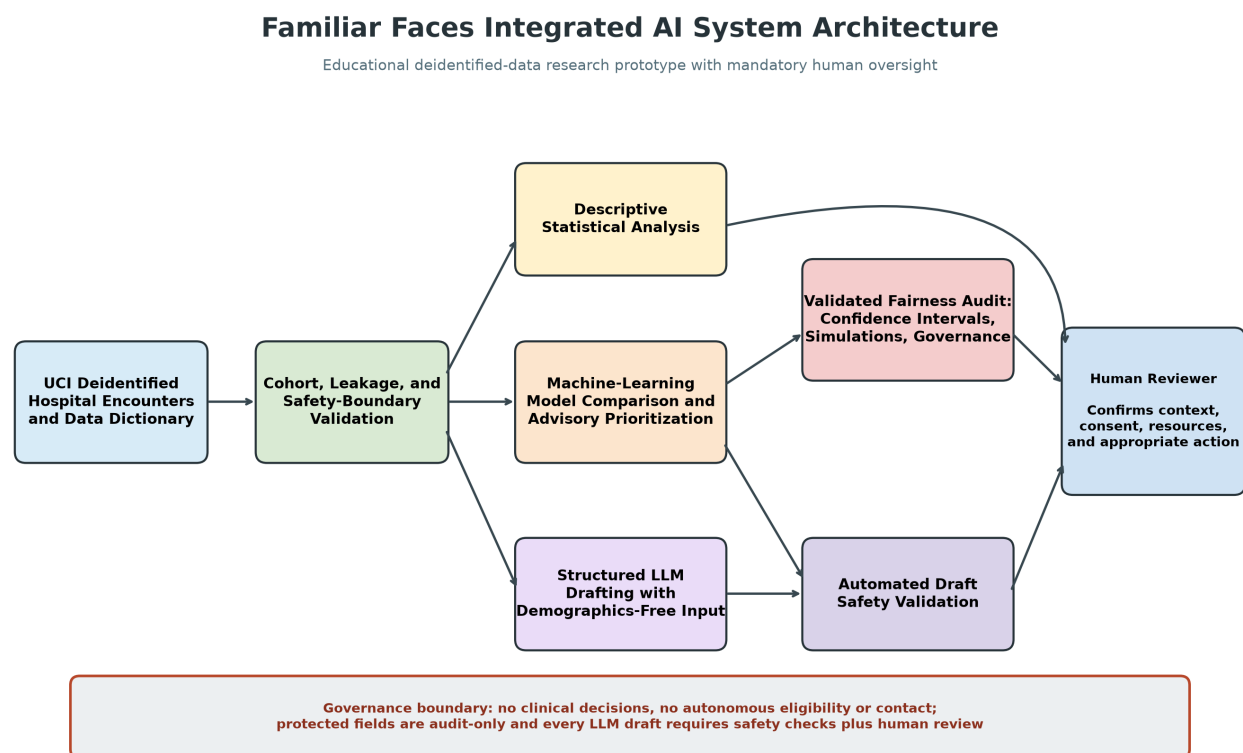
Deep Learning Systems and Agentic AI insights. These subjects influenced design choices but are not misrepresented as implemented domains. Sequential deep models were unnecessary for the selected tabular encounter features, where interpretability and limited evidence favored simpler classifiers. Likewise, a fixed, human-supervised workflow does not need an autonomous agent. Agentic AI work contributed the stop conditions, permission mindset, and escalation pattern. The three implemented domains are data analysis, machine learning, and generative AI; the other subjects supply justified constraints.

3. System Architecture

The architecture separates cohort preparation, descriptive analysis, model comparison, protected-group auditing, LLM drafting, deterministic validation, and human review. Age, gender, and race follow an audit-only path and never enter the predictive or generative feature sets. The model probability can select records for draft preparation, but neither the model nor the LLM can initiate action. This separation keeps prediction, auditing, language generation, and accountability inspectable rather than collapsing them into an opaque score.

Figure 1

Familiar Faces Human-Supervised System Architecture



Note. Protected attributes are audit-only. LLM inputs are deidentified and demographics-free. No component authorizes contact or clinical action.

4. Results and Evaluation

Table 1

Integrated Prototype Results

System metric	Observed result
Eligible unique-patient encounters	69,990
Held-out encounters	17,498
Selected model	Histogram Gradient Boosting
ROC AUC (95% CI)	0.641 [0.626, 0.656]
Average precision (95% CI)	0.171 [0.156, 0.186]
Test precision	0.000
Test recall	0.000
Fairness review triggers	0
Null-simulation trigger rate	0.0%
Injected-disparity detection	100.0%
LLM drafts passing safety gate	5 of 5

The real outcome makes model comparison empirical. Training-only cross-validation selected histogram gradient boosting by the predeclared average-precision rule, with ROC AUC used only as a secondary tie-breaker. The selected model was then evaluated once on the final test data. Random forest and histogram gradient boosting were effectively tied in cross-validated average precision (0.168727 versus 0.168709), so the numerical winner should not be interpreted as evidence of meaningful superiority. Compared with the earlier test-set selection workflow, average precision increased only slightly, from 0.164 to 0.171. The substantive improvement is methodological credibility, not materially stronger discrimination. The result shows some ranking information beyond the 0.090 base rate, but it does not establish calibration, transportability, treatment effect, or outreach benefit.

5. Fairness Validation and Governance

Removing protected attributes from training does not guarantee equitable behavior because other features and the outcome can encode structural differences. Health-management algorithms have produced racial disparity even when designers used a seemingly reasonable proxy objective (Obermeyer et al., 2019). Familiar Faces therefore retains protected fields only for post-model auditing. Each group rate includes a 95% Wilson interval. A comparison requires at least 500 group records and at least 200 observations in the relevant outcome denominator. Each eligible group is compared with the largest reference group.

A review trigger requires both practical and statistical evidence: an absolute rate difference of at least 0.10 and a Holm-adjusted p-value below .05 across the complete comparison family. The method itself is tested rather than assumed valid. In each null simulation, three independent audit fields are randomly assigned and flag, true-positive, and false-positive rates are evaluated. The observed familywise false-positive rate was 0.0%. In the matched injected-disparity experiment, a 0.15 review-flag difference was detected in 100.0% of runs. These experiments address false alarms and power under stated conditions; they do not certify fairness in all settings.

The real-data audit produced 0 REVIEW_REQUIRED comparisons, including age-related flag and error-rate differences and one race-related true-positive-rate difference. These are governance signals, not proof of discrimination or permission to alter services automatically. They require contextual investigation, stakeholder review, sensitivity analysis, and potential model revision. NIST's AI Risk Management Framework similarly treats risk work as continuing governance, mapping, measurement, and management rather than a one-time technical score (Tabassi, 2023).

6. Ethics, Constraints, and Responsible Use

The primary ethical risk is unequal allocation of staff attention. Additional risks include automation bias, privacy misuse, historical-data drift, unsupported LLM inferences, and purpose expansion from review support to eligibility control. The system responds with layered safeguards: deidentified public research data; audit-only protected attributes; patient-independent evaluation; explicit uncertainty; constrained structured generation; a separate deterministic text gate; no autonomous action; and mandatory human review. Human oversight is not treated as a slogan: an authorized reviewer must verify facts, consent, current context, available resources, and the appropriateness of any next step.

Important limitations remain. The data are from 1999–2008, involve hospitalized patients with diabetes, and do not represent a modern community-paramedicine population. One encounter per patient improves independence but discards later history. Readmission may be planned, unavoidable, or unrelated to appropriate outreach. The models have not been externally validated or prospectively tested. The fairness simulations use controlled binary assignments and one injected effect size. LLM safety checks cannot detect every misleading implication. The prototype has not undergone clinical, legal, privacy, cybersecurity, accessibility, usability, or organizational validation.

Before any real-world research pilot, an organization would need an approved protocol, lawful and authorized data access, privacy and security controls, community and patient participation, independent model validation, calibration and subgroup analysis on representative data, reviewer training, audit logging, appeal and escalation procedures, incident ownership, and continuous monitoring. Prospective evaluation would need patient-centered and program outcomes—not just readmission prediction—and a comparison showing whether the AI-assisted workflow improves decisions without worsening inequity.

7. Professional Reflection and Conclusion

This revision sharpened the difference between assembling components and integrating valid evidence. A model that predicts its own label-generating formula is reproducible but not informative. A fairness threshold without false-positive and detection testing can create confidence without validity. A text template can demonstrate safeguards but is not generative AI. Replacing those shortcuts with a real outcome, a validated audit, and executed LLM inference made the portfolio artifact more honest and technically meaningful.

Familiar Faces now demonstrates system-level judgment across data analysis, machine learning, generative AI, fairness evaluation, and governance. Its value does not depend on claiming deployment readiness. The strongest conclusion is that the integrated workflow runs, exposes its limitations, and stops at the correct boundary. Further work should begin with stakeholder and governance review, not operational use.

References

- Clore, J., Cios, K., DeShazo, J., & Strack, B. (2014). *Diabetes 130-US Hospitals for Years 1999-2008* [Data set]. UCI Machine Learning Repository. <https://doi.org/10.24432/C5230J>
- Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019). Dissecting racial bias in an algorithm used to manage the health of populations. *Science*, 366(6464), 447-453. <https://doi.org/10.1126/science.aax2342>
- OpenAI. (2026). *Structured model outputs*. <https://developers.openai.com/api/docs/guides/structured-outputs>
- Strack, B., DeShazo, J. P., Gennings, C., Olmo, J. L., Ventura, S., Cios, K. J., & Clore, J. N. (2014). Impact of HbA1c measurement on hospital readmission rates: Analysis of 70,000 clinical database patient records. *BioMed Research International*, 2014, Article 781670. <https://doi.org/10.1155/2014/781670>
- Tabassi, E. (2023). *Artificial intelligence risk management framework (AI RMF 1.0)* (NIST AI 100-1). National Institute of Standards and Technology. <https://doi.org/10.6028/NIST.AI.100-1>